AnalogEdge

PLL Behavioral Modeling and Verification

Lock-time analysis, jitter sensitivity and integration guidelines

Persevex VLSI Course - Project 1

Domain: VLSI (mixed-signal clocking)

Report date: October 2026

1. Introduction

A phase-locked loop (PLL) generates a fast, clean clock from a slower reference. Clock quality (how quickly the PLL locks, how it reacts to disturbances and how much timing jitter it adds) directly limits the performance of digital and mixed-signal systems. This project builds a behavioral model of a charge-pump PLL for the AnalogEdge case study, simulates its lock behavior, studies how loop bandwidth and damping affect lock time and jitter, and turns the findings into integration guidelines.

The work follows the four phases of the project brief:

- **Phase 1 - Block modeling:** phase-frequency detector (PFD), charge pump, loop filter, VCO and divider; target frequency and lock condition.
- **Phase 2 - Dynamic simulation:** lock time, frequency settling and phase error under perturbations.
- **Phase 3 - Noise and jitter sensitivity:** parameter sweeps of loop bandwidth and damping.
- **Phase 4 - Integration guidelines:** clocking recommendations, model limitations and assumptions.

Deliverables covered by this report: PLL behavioral model files (SourceCode folder), lock and jitter characterization plots (Plots folder and Sections 5 to 7), and the integration and design recommendations (Section 8).

2. PLL Architecture and Block Models

The PLL is a negative-feedback loop. The divided VCO output is compared with the reference; the error steers the VCO until both signals agree in frequency and phase.

Reference (10 MHz) > PFD > Charge pump > Loop filter (R + C) > VCO (100 MHz output) > Divide-by-N (back to PFD)

Block	Function	Behavioral model used
PFD (tri-state)	Detects which clock edge arrives first and by how much	Phase error e (in reference cycles) = reference phase - VCO phase / N; pulse width proportional to |e|, limited to one cycle
Charge pump	Converts the UP/DOWN pulses into current	Charge per cycle Q = Icp x Tref x e (positive for UP, negative for DOWN)
Loop filter (series R, C)	Integrates the charge and adds a proportional (damping) term	Vc = Vc + Q / C (integral path); VCO phase kick = Kvco x R x Q (proportional path)
VCO	Output frequency set by control voltage	f_{vco} = f0 + Kvco x (Vc - Vc0)
Divider (divide by N)	Brings the VCO frequency down to the reference frequency	Feedback phase = VCO phase / N

Target and lock condition. The target output is 100 MHz (N = 10, f_{ref} = 10 MHz). The loop is declared **locked** when the absolute phase error stays below 0.3 ns for 30 consecutive reference cycles (0.3 ns is 0.3% of the 100 ns reference period). Because the VCO frequency is tied to the phase, a locked loop also has the correct output frequency.

3. Design Parameters

A single baseline design is used throughout. For a second-order charge-pump PLL with a series R-C filter, the natural frequency and damping factor follow from the component values:

$$\omega_n = sqrt(Icp \times Kvco / (2\pi \times N \times C)), \quad \zeta = R \times C \times \omega_n / 2$$

(Kvco in rad/s per volt inside this formula.) The resistor is chosen for the desired damping factor: $R = 2\zeta / (\omega_n \times C)$.

Parameter	Value	Parameter	Value
Reference frequency	10 MHz	Charge pump current Icp	100 uA
Divider N	10	VCO gain Kvco	50 MHz/V
Output frequency	100 MHz	VCO centre (Vc0 = 1.0 V)	100 MHz
Loop filter C	100 pF	Loop filter R	6.26 kOhm
Damping factor	0.7	Natural frequency	356 kHz
Unity-gain crossover	549 kHz (about $f_{ref}/18$)	Phase margin	65 degrees

The VCO deliberately starts at 90 MHz (Vc = 0.8 V), a 10% frequency error, so that the simulation shows full lock acquisition.

4. Simulation Methodology

Time-domain behavioral model (Python). The loop is updated once per reference cycle, which is exact for a tri-state PFD with a charge pump because the correction in each cycle is delivered as one packet of charge. In every cycle: (1) the phase error is computed, (2) the charge-pump packet Q is formed, (3) the capacitor voltage and the VCO phase are updated, and (4) the VCO runs at the new frequency until the next reference edge. This keeps the model fast and lets it reproduce lock acquisition, re-locking and perturbation response.

Scenario simulated (80 microseconds):

- 0 us: start with the VCO at 90 MHz (lock acquisition).
- 30 us: reference frequency steps from 10 MHz to 10.5 MHz (target moves to 105 MHz).
- 50 us to 55 us: a +20 mV disturbance is added to the VCO control voltage (supply or substrate noise).

Linear model for sweeps (Python, scipy). The closed loop is a second-order transfer function, $H(s) = (2\zeta\omega_n s + \omega_n^2) / (s^2 + 2\zeta\omega_n s + \omega_n^2)$. It is used for fast sweeps of loop bandwidth and damping (settling time, overshoot, phase margin) and for the jitter integration in Section 7.

SystemVerilog model. An equivalent event-driven SystemVerilog model (design.sv and testbench.sv) is included as supplementary source code. The results in this report come from the Python models.

5. Results: Lock Behavior and Perturbations (Phase 2)

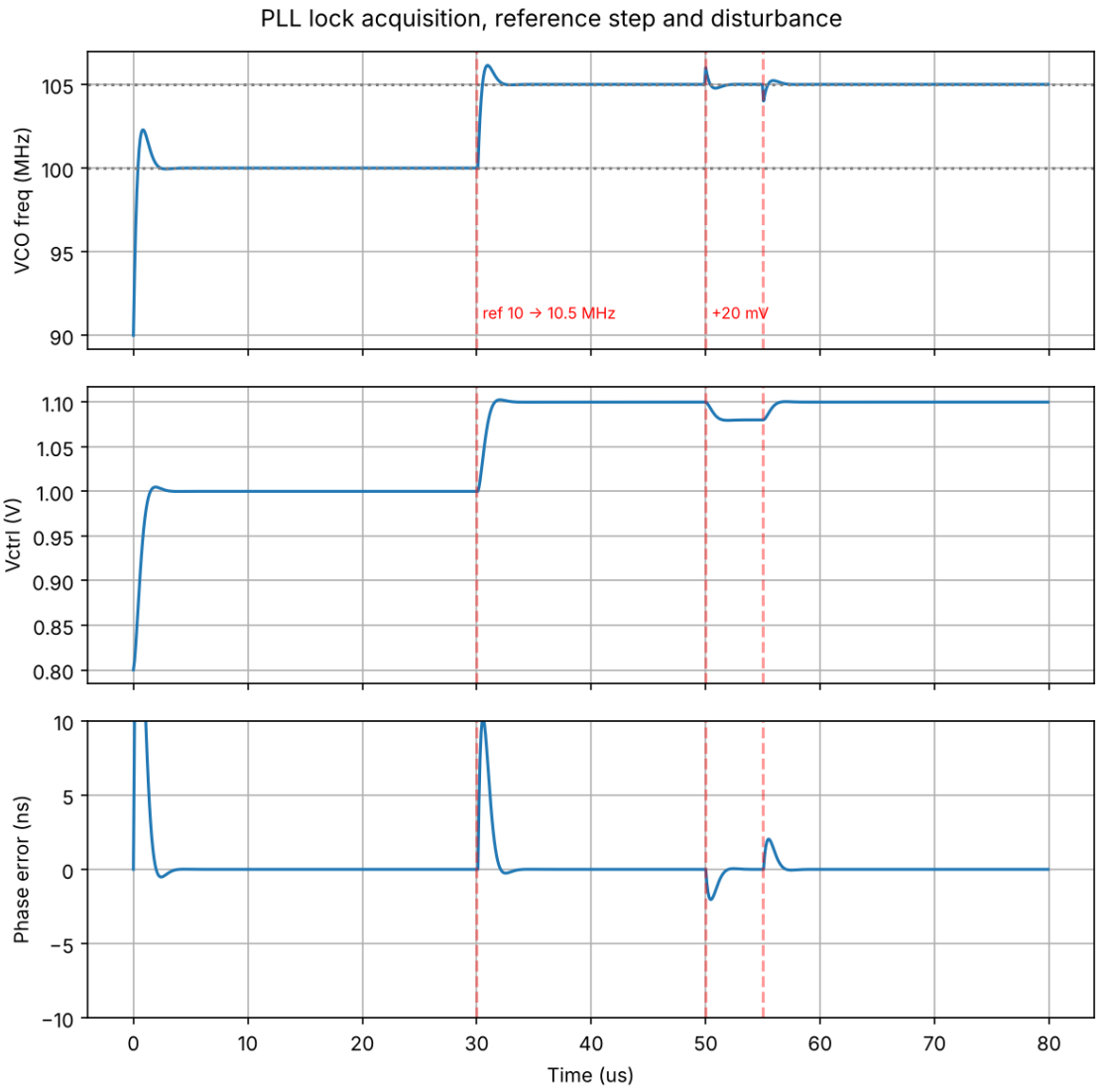


Figure 1. VCO frequency, control voltage and phase error for lock acquisition, a reference step at 30 us and a 20 mV disturbance between 50 and 55 us.

Event	Observation
Lock acquisition (90 MHz to 100 MHz)	VCO crosses 100 MHz at 0.4 us, peaks at 102.3 MHz (22.8% overshoot of the 10 MHz step), phase lock declared at 3.0 us; Vc settles at 1.00 V
Reference step 10 to 10.5 MHz (30 us)	Peak frequency 106.1 MHz (22.8% overshoot of the 5 MHz step); re-locks at 32.0 us (about 2.0 us after the step); peak phase error about 10 ns; Vc moves from 1.00 V to 1.10 V; output settles at 105.00 MHz
+20 mV disturbance applied (50 us)	Frequency kicked by 1.0 MHz (20 mV x 50 MHz/V); peak phase error about 2.0 ns; lock regained at 51.5 us (about 1.5 us); loop pulls Vc down by 20 mV to cancel it
Disturbance removed (55 us)	Equal and opposite 1.0 MHz transient (frequency dips to 104.0 MHz); peak phase error about 2.0 ns; lock regained at 56.6 us (about 1.6 us)

Three ways to quote lock time. The numbers below depend on the definition, so the definition must be stated. With the baseline design: phase lock (|e| below 0.3 ns for 30 cycles) occurs at 3.0 us; the

frequency stays within +/-1 MHz of the target after 1.6 us; the linear model's 1% settling time of the frequency step response is 2.3 us. All three agree that the baseline PLL locks within a few microseconds.

Model cross-check. The time-domain simulation shows a frequency overshoot of 22.8% for both the acquisition and the reference step, close to the 21% predicted by the linear second-order model for a damping factor of 0.7. The two independent models therefore agree.

Sensitivity to disturbances. A 20 mV disturbance shifts the output by 1 MHz (1% of 100 MHz). The PLL rejects it within about 1.5 us, but any noise on the control line inside the loop bandwidth appears as output jitter, which motivates the supply-isolation recommendations in Section 8.

6. Parameter Sweeps: Bandwidth and Damping (Phase 3)

6.1 Loop bandwidth

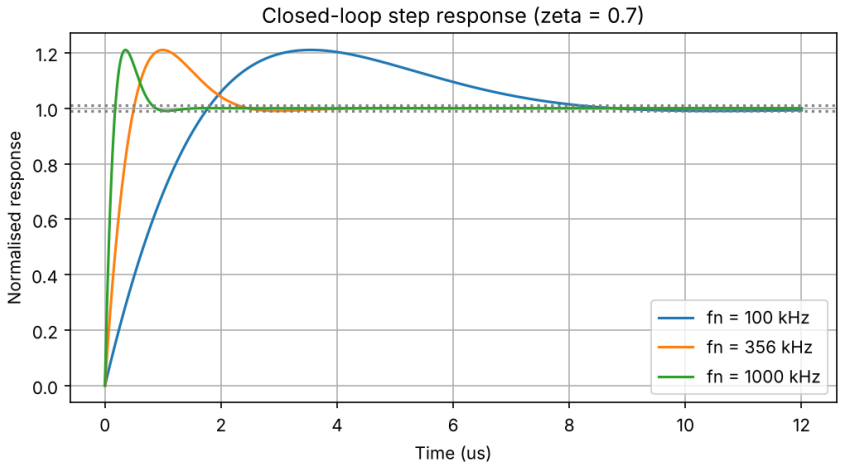


Figure 2. Closed-loop step response of the linear model for three loop bandwidths (damping 0.7). Dotted lines mark the +/-1% settling band.

Natural frequency	Settling time (1%, linear model)	Settle to +/-1 MHz (time-domain sim)	Overshoot	Phase margin
100 kHz	8.18 us	5.80 us	21%	65 deg
200 kHz	4.09 us	2.90 us	21%	65 deg
356 kHz (baseline)	2.30 us	1.60 us	21%	65 deg
600 kHz	1.36 us	0.90 us	21%	65 deg
1 MHz	0.82 us	0.50 us	21%	65 deg

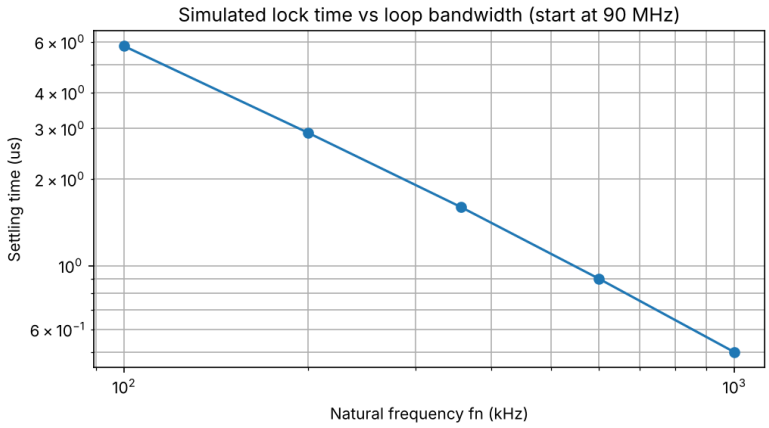


Figure 3. Lock time against loop bandwidth from the time-domain simulation (start at 90 MHz).

Lock time is inversely proportional to loop bandwidth: doubling the bandwidth halves it. With damping fixed, overshoot and phase margin stay constant in the continuous-time model, so bandwidth alone does not change the shape of the response. The time-domain values are shorter than the linear 1% settling times because they use a wider band: +/-1 MHz is 10% of the 10 MHz initial error, while the linear 1% band is 1% of the step.

6.2 Damping factor

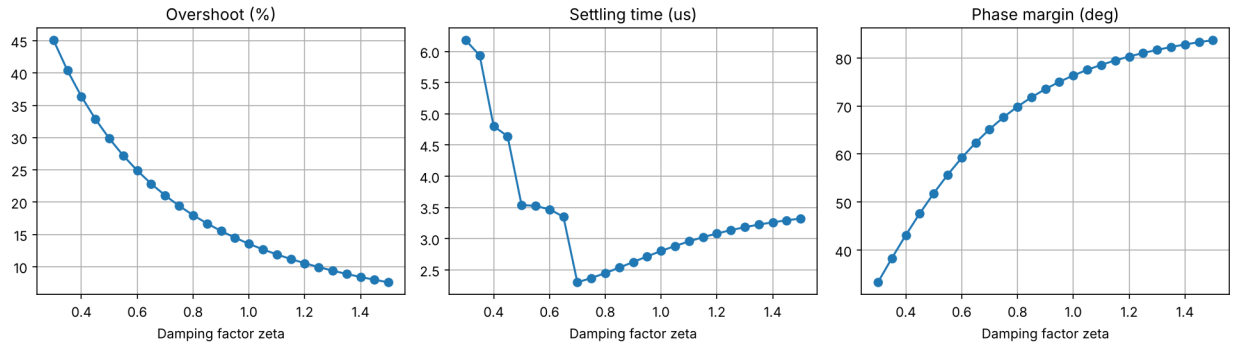


Figure 4. Overshoot, settling time and phase margin against damping factor (natural frequency 356 kHz).

Damping	Overshoot	Settling time (1%)	Phase margin	Total RMS jitter
0.3	45.1%	6.18 us	33 deg	20.6 ps
0.5	29.8%	3.54 us	52 deg	16.4 ps
0.7 (baseline)	21.0%	2.30 us	65 deg	14.4 ps
1.0	13.5%	2.80 us	76 deg	12.9 ps
1.5	7.6%	3.32 us	84 deg	12.1 ps

Low damping rings and has poor phase margin (33 degrees at 0.3). High damping removes overshoot and gives large margin but slows settling. The fastest settling is at a damping factor of about 0.7, and values from 0.7 to 1.0 give a good balance of speed (2.3 to 2.8 us) and margin (65 to 76 degrees).

7. Noise and Jitter Sensitivity (Phase 3)

Noise model. Two independent phase-noise sources are considered, using assumed (not measured) levels: (a) reference noise, white at -130 dBc/Hz, which reaches the output multiplied by N^2 (+20 dB for N = 10) and filtered by the low-pass closed-loop response H(s); (b) VCO noise of -110 dBc/Hz at 1 MHz offset with a $1/f^2$ slope, filtered by the high-pass response 1 - H(s). The two contributions are integrated from 1 kHz to $f_{ref}/2$ and converted to RMS time jitter at 100 MHz.

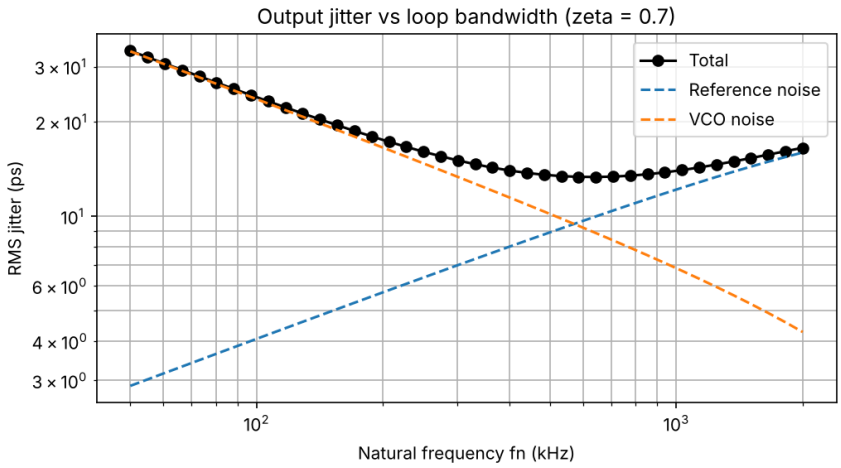


Figure 5. RMS output jitter against loop bandwidth (damping 0.7): reference-noise and VCO-noise contributions and their total.

Natural frequency	Total RMS jitter	Dominant contribution
100 kHz	24.0 ps	VCO noise (loop too slow to suppress it)
200 kHz	17.5 ps	VCO noise
356 kHz (baseline)	14.4 ps	Balanced
600 kHz	13.4 ps	Balanced (minimum near 643 kHz, 13.35 ps)
1 MHz	14.0 ps	Reference noise (loop passes more of it)

Trade-off. A wider loop suppresses VCO noise and locks faster, but passes and amplifies more reference noise and reduces stability margin in a real sampled loop. Jitter therefore has a minimum, here near 640 kHz (about $f_{ref}/15$). At 1 MHz natural frequency the unity-gain crossover is about 1.5 MHz ($f_{ref}/6.5$), beyond the usual rule of thumb that the crossover stay below $f_{ref}/10$. The continuous-time model is optimistic there, because the sampling delay of the PFD is not included, so very wide loops should not be trusted.

8. Integration Guidelines (Phase 4)

Topic	Recommendation	Reason from this study
Loop bandwidth	Natural frequency between $f_{ref}/30$ and $f_{ref}/15$ (about 330 to 670 kHz for 10 MHz); crossover below $f_{ref}/10$	Jitter minimum near 640 kHz; lock in 1 to 3 us; model stays valid
Damping factor	0.7 to 1.0	Fastest settling at 0.7; phase margin 65 to 76 deg; overshoot 13 to 21%
Reference clock	Use a low-jitter, clean reference (crystal or low-noise oscillator)	Reference noise is multiplied by N^2 and dominates at wide bandwidth
Divider ratio	Keep N as small as the application allows	Each doubling of N adds 6 dB of reference-noise gain at the output
VCO control line (Vc)	Short, shielded routing; guard ring; no digital signals nearby; keep the filter capacitor close to the VCO	20 mV on Vc shifts the output by 1 MHz
Supply and substrate	Dedicated, regulated and filtered supply for the VCO and charge pump; separate analog and digital grounds joined at one point	Supply noise couples into Vc and appears as jitter
Charge pump	Match UP and DOWN currents; minimise leakage; minimise pulse overlap	Mismatch and leakage create static phase error and reference spurs (not modeled here)
Extra filtering	Add a small second capacitor (about C/10 to C/20) for reference-spur suppression, then re-check phase margin	The model has only the R-C filter
Start-up and lock monitor	Use a lock detector with the same rule as Section 2 and hold downstream logic in reset until lock; allow at least 5 us at the baseline design	Lock streak starts at 3 us; re-lock after disturbances takes 1.5 to 2 us

9. Model Limitations and Assumptions

- **Ideal charge pump:** equal UP and DOWN currents, no leakage, no mismatch, no dead zone.
- **Linear behavioral VCO:** constant gain Kvco over the range, no tuning non-linearity; control voltage clamped to 0 to 2 V.
- **Integer-N only:** no fractional-N or delta-sigma noise.
- **Second-order loop:** only the series R-C filter is modeled; no second capacitor and no higher-order poles.
- **Per-cycle sampling:** the time-domain model captures behavior at the reference rate; it does not show ripple within a reference cycle or reference spurs.
- **Assumed noise levels:** the jitter results use assumed phase-noise numbers (-130 dBc/Hz reference, -110 dBc/Hz at 1 MHz for the VCO). Absolute picoseconds will change with real silicon data; the trends and the location of the optimum are the useful result.
- **Linear jitter analysis:** jitter comes from frequency-domain integration, not from a noisy transient simulation.
- **No process, voltage and temperature variation:** a corner analysis is needed before use in a product.

10. Conclusion

A charge-pump PLL (10 MHz reference, divide-by-10, 100 MHz output) was modeled block by block and simulated. The baseline design (natural frequency 356 kHz, damping 0.7, phase margin 65 degrees) locks from a 10% frequency error in about 3 microseconds, re-locks in about 2 microseconds after a reference step, and recovers from a 20 mV control-voltage disturbance in about 1.5 microseconds. Sweeps show that lock time scales inversely with loop bandwidth, that damping between 0.7 and 1.0 balances speed and stability, and that output jitter has a minimum near 640 kHz where VCO-noise suppression and reference-noise pass-through balance. The integration guidelines translate these results into bandwidth, damping, reference-clock, layout and supply recommendations, and the limitations list shows what must be added (charge-pump non-idealities, second-order filtering, corners) before using the design in a product.

Appendix: files in the project folder

File	Purpose
SourceCode/pll_sim.py	Time-domain behavioral simulation (lock, reference step, disturbance, bandwidth check)
SourceCode/pll_analysis.py	Linear model: lock-time, damping and jitter sweeps
SourceCode/design.sv, testbench.sv	Supplementary SystemVerilog behavioral model and testbench
Plots/	All figures used in this report
Results/	Printed output of the two Python programs